

Task No.: 3**Date: 05-01-2026**

Title: Develop an Android application to draw basic graphical primitives on the screen

Aim:

To create an Android application that uses the Canvas and Paint classes to draw basic graphical primitives on the screen.

Code:

MainActivity.kt

```
package com.example.madtask4
```

```
import android.os.Bundle
```

```
import androidx.appcompat.app.AppCompatActivity
```

```
class MainActivity : AppCompatActivity() {
```

```
    override fun onCreate(savedInstanceState: Bundle?) {
```

```
        super.onCreate(savedInstanceState)
```

```
        setContentView(R.layout.activity_main)
```

```
    }
```

```
}
```

FourQuadrantView.kt

```
package com.example.madtask4
```

```
import android.content.Context
```

```
import android.graphics.*
```

```
import android.util.AttributeSet
```

```
import android.view.MotionEvent
```

```
import android.view.View
```

```
class FourQuadrantView @JvmOverloads constructor(
```

```
    context: Context,
```

```
    attrs: AttributeSet? = null
```

```

) : View(context, attrs) {

    private val shapePaint = Paint().apply {
        isAntiAlias = true
        strokeWidth = 5f
        style = Paint.Style.STROKE
        color = Color.BLACK
    }

    private val textPaint = Paint().apply {
        isAntiAlias = true
        color = Color.BLACK
        textSize = 36f
        textAlign = Paint.Align.CENTER
        style = Paint.Style.FILL
    }

    private val fillColors = arrayOf(
        Color.RED,
        Color.GREEN,
        Color.BLUE
    )

    private var circleState = 0
    private var rectState = 0
    private var ellipseState = 0
    private var triangleState = 0

    override fun onDraw(canvas: Canvas) {
        super.onDraw(canvas)

        val w = width.toFloat()
        val h = height.toFloat()
        val halfW = w / 2
        val halfH = h / 2

        applyStyle(circleState)
        val cx = halfW / 2
        val cy = halfH / 2
        val radius = Math.min(halfW, halfH) / 4
        canvas.drawCircle(cx, cy, radius, shapePaint)
    }
}

```

```
canvas.drawText("Circle", cx, cy + radius + 50, textPaint)
```

```
applyStyle(rectState)
```

```
val rectLeft = halfW + halfW / 4
```

```
val rectTop = halfH / 4
```

```
val rectRight = halfW + 3 * halfW / 4
```

```
val rectBottom = 3 * halfH / 4
```

```
canvas.drawRect(rectLeft, rectTop, rectRight, rectBottom, shapePaint)
```

```
canvas.drawText("Rectangle", halfW + halfW / 2, rectBottom + 50, textPaint)
```

```
applyStyle(ellipseState)
```

```
val oval = RectF(
```

```
    halfW / 4,
```

```
    halfH + halfH / 4,
```

```
    3 * halfW / 4,
```

```
    halfH + 3 * halfH / 4
```

```
)
```

```
canvas.drawOval(oval, shapePaint)
```

```
canvas.drawText("Ellipse", halfW / 2, oval.bottom + 50, textPaint)
```

```
applyStyle(triangleState)
```

```
val path = Path()
```

```
path.moveTo(halfW + halfW / 2, halfH + halfH / 4)
```

```
path.lineTo(halfW + halfW / 4, halfH + 3 * halfH / 4)
```

```
path.lineTo(halfW + 3 * halfW / 4, halfH + 3 * halfH / 4)
```

```
path.close()
```

```
canvas.drawPath(path, shapePaint)
```

```
canvas.drawText(
```

```
    "Triangle",
```

```
    halfW + halfW / 2,
```

```
    halfH + 3 * halfH / 4 + 50,
```

```
    textPaint
```

```
)
```

```
}
```

```
private fun applyStyle(state: Int) {
```

```
    if (state == 0) {
```

```
        shapePaint.style = Paint.Style.STROKE
```

```
        shapePaint.color = Color.BLACK
```

```
    } else {
```

```
        shapePaint.style = Paint.Style.FILL_AND_STROKE
```

```

        shapePaint.color = fillColors[state - 1]
    }
}

override fun onTouchEvent(event: MotionEvent): Boolean {
    if (event.action == MotionEvent.ACTION_DOWN) {

        val x = event.x
        val y = event.y
        val halfW = width / 2f
        val halfH = height / 2f

        when {
            x < halfW && y < halfH ->
                circleState = (circleState + 1) % 4

            x > halfW && y < halfH ->
                rectState = (rectState + 1) % 4

            x < halfW && y > halfH ->
                ellipseState = (ellipseState + 1) % 4

            x > halfW && y > halfH ->
                triangleState = (triangleState + 1) % 4
        }

        invalidate()
    }
    return true
}
}

```

activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="@android:color/white">

```

```
<TextView
    android:id="@+id/headerText"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="@string/header_text"
    android:textSize="18sp"
    android:textStyle="bold"
    android:textColor="@android:color/black"
    android:gravity="center"
    android:padding="16dp" />

<FrameLayout
    android:layout_width="match_parent"
    android:layout_height="0dp"
    android:layout_weight="1">

    <com.example.madtask4.FourQuadrantView
        android:layout_width="match_parent"
        android:layout_height="match_parent" />

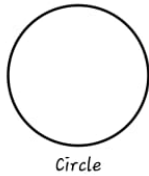
</FrameLayout>

</LinearLayout>
```

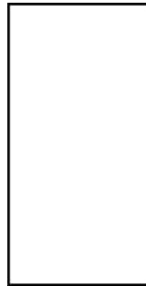
Output:

Name: Yoga Shri P.K
Roll No: 23BCS192

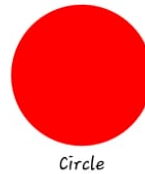
Name: Yoga Shri P.K
Roll No: 23BCS192



Circle



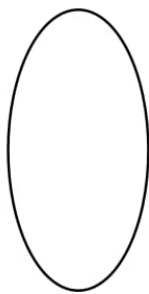
Rectangle



Circle



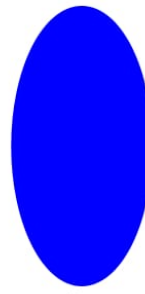
Rectangle



Ellipse



Triangle



Ellipse



Triangle

Result:

An android application using the paint and canvas classes are imported and basic diagrams are drawn and executed successfully.